COMPARISON OF TWO BIG PRO-ETALE SITES WITH ONE CONTAINED IN THE OTHER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the verification of the five conditions of the bigger-site criterion for the absolute and relative inclusions, in particular the refinement of pro-etale coverings and the compatibility of fibre products with those of schemes.

Editorial note. The label lemma-change-alpha names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-change-alpha –start 6591 –end 6633 –local-id 28010 returned RESULT=OK (L28010 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Criterion for the inclusion of a site to induce morphisms of topoi with a section. Sites, the lemma listing the conditions (a)-(e) on the inclusion functor under which the induced morphisms of topoi exist with f composed with g isomorphic to the identity. Supplies the criterion whose five conditions are verified in the proof.
- Refinement of coverings for the inclusion of big pro-etale sites. Same chapter, the lemma that every pro-etale covering of an object of the big pro-etale site is refined by a covering of the site. Supplies condition (a) of the criterion for both inclusions.
- Fibre products in the big pro-etale sites agree with those of schemes. Same chapter, the lemma that the big pro-etale sites have fibre products agreeing with fibre products in the category of schemes. Supplies condition (d): fibre products exist and are compatible with those of schemes, hence are preserved by the inclusion.

Source reference. Original reference 098G: sources/stacks-proetale/pinned-original-source.tex, source lines 2465–2465. Prerequisite pointer invoked at proof line(s) 6594; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 098J: sources/stacks-proetale/pinned-original-source.tex, source lines 6516–6516. Prerequisite pointer invoked at proof line(s) 6629; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00XU: sources/stacks-chow/tags.txt, source lines 1232–1232. Prerequisite pointer invoked at proof line(s) 6597; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Suppose given big sites $Sch_{pro\text{-}étale}$ and $Sch'_{pro\text{-}étale}$ as in Definition 098G. Assume that $Sch_{pro\text{-}étale}$ is contained in $Sch'_{pro\text{-}étale}$. The inclusion functor $Sch_{pro\text{-}étale} \to Sch'_{pro\text{-}étale}$ satisfies the assumptions of Sites, Lemma 00XU. There are morphisms of topoi*

$$\begin{aligned} g : Sh(Sch_{pro\text{-}étale}) &\longrightarrow Sh(Sch'_{pro\text{-}étale}) \\ f : Sh(Sch'_{pro\text{-}étale}) &\longrightarrow Sh(Sch_{pro\text{-}étale}) \end{aligned}$$

such that $f \circ g \cong id$. For any object S of $Sch_{pro\text{-}étale}$ the inclusion functor $(Sch/S)_{pro\text{-}étale} \to (Sch'/S)_{pro\text{-}étale}$ satisfies the assumptions of Sites, Lemma 00XU also. Hence similarly we obtain morphisms

$$\begin{aligned} g : Sh((Sch/S)_{pro\text{-}étale}) &\longrightarrow Sh((Sch'/S)_{pro\text{-}étale}) \\ f : Sh((Sch'/S)_{pro\text{-}étale}) &\longrightarrow Sh((Sch/S)_{pro\text{-}étale}) \end{aligned}$$

with $f \circ g \cong id$.

Proof. Assumptions (b), (c), and (e) of Sites, Lemma 00XU are immediate for the functors $Sch_{pro\text{-}étale} \to Sch'_{pro\text{-}étale}$ and $(Sch/S)_{pro\text{-}étale} \to (Sch'/S)_{pro\text{-}étale}$. Property (a) holds by Lemma 098J. Property (d) holds because fibre products in the categories $Sch_{pro\text{-}étale}$, $Sch'_{pro\text{-}étale}$ exist and are compatible with fibre products in the category of schemes. □